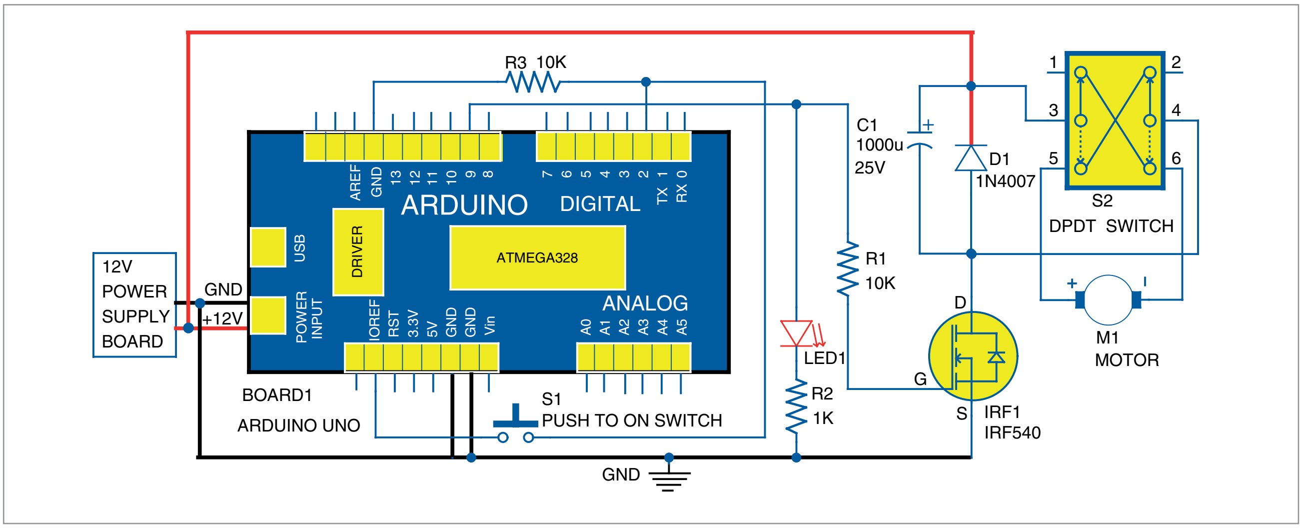

Fig. 4: Circuit diagram of the time based speed control switch for DC motor

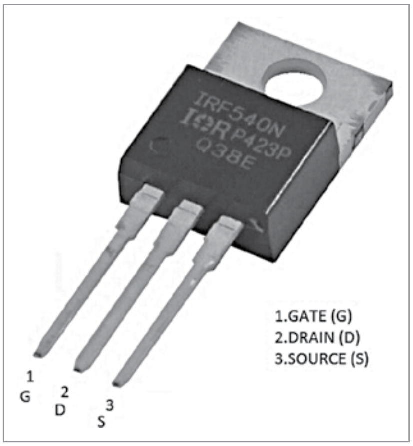

Fig. 5: Pin details of MOSFET IRF540

PARTS LIST

Semiconductors:	
Board1	- Arduino Uno board
T1	- IRF540 MOSFET
D1	- 1N4007 rectifier diode
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1, R3	- 10-kilo-ohms
R2	- 1-kilo-ohms
Capacitor:	
C1	- 1000µF, 25V electrolytics
Miscellaneous:	
S1	- Pushbutton switch
S2	- 2A, DPDT switch
Motor	- 12V, 500mA DC motor
	- 12V, 1A DC power supply with 5.5mm x 2.1mm, centre-positive male barrel jack
	- Heat-sink for T1

The pin-outs of IRF540 are shown in Fig. 5. The drain is connected to 12V DC supply via diode D1, which protects the circuit during motor power on and off conditions. Capacitor C1 (1000µF, 25V) acts as a smoothing device. DPDT switch S2 connected between diode D1 and motor is used as a forward-reverse switch.

For the prototype, a small 12V DC motor having 500mA rating and a 2A DPDT switch were used. A 12V, 1A power supply was used for the prototype. The ratings of 12V DC power supply, MOSFET, diode D1, and DPDT switch S2 are decided based on ratings of the DC motor.

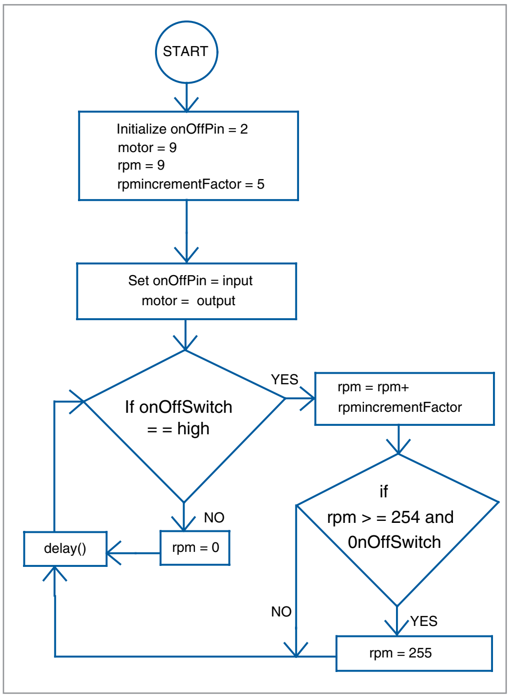

Fig. 6: Flowchart of the program

Software

The circuit uses software program *Time2SpeedSwitch.ino* written in Arduino programming language that is loaded into the internal memory of Arduino Uno. ATmega328P on Arduino Uno comes with a pre-programmed boot loader that allows users to upload a new code to it without using a peripheral hardware programmer. Arduino IDE 1.6.4 is used to compile and upload the program. The sketch (software) is the heart of the system and carries out all the main functions.

The program is simple and easy to understand. Comments are given at the end of each command line. Input and output pins are initialised. The *rpmIncrementFactor* is a numeric value representing the rate of increase of the motor speed in steps. The value is user-defined and can be set as per requirement.

The *delay( )* function determines the rate at which the speed of the motor increases. For increasing speed, decrease the delay value, and vice versa. For better understanding of the program code its flowchart is shown in Fig. 6.

EFY Note

The source code of this project is available for free download at *source.efymag.com*